

Measurement of PM_{2.5} emission potential from soil using the UC Davis resuspension test chamber

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Abstract

Human health effects have been linked to airborne concentrations of fine particulate matter. One source of fine particulate matter in the atmosphere is resuspended soil dust from a variety of activities, including agricultural operations. We have established a method to measure the potential of soil to emit fugitive dust in the PM₁₀ or PM_{2.5} size range. The method is repeatable, and provides an index of PM₁₀ or PM_{2.5} dust that is highly correlated to the soil texture. The ratio of the PM_{2.5} Index to the PM₁₀ Index produced by this method is similar to field observations of ambient PM_{2.5} and PM₁₀ concentrations downwind of agricultural operations in the San Joaquin Valley of California. The PM_{2.5} or PM₁₀ Index will be a more useful parameter to estimate the potential of a soil to emit fugitive dust than the currently used dry silt content of soil. Research is currently underway to relate the PM₁₀ and PM_{2.5} Index to measured emission factors, accounting for soil moisture and type of agricultural operation, so that a more reliable predictive equation can be developed for agricultural practices.

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1. Introduction

Human health effects, including increased mortality and morbidity, have been linked to exposure to airborne particulate matter (Pope and Dockery, 1998; Pope et al., 1995; U.S. EPA, 1996). The health effects are more strongly correlated to fine particulate matter, those particles less than or equal to 2.5 μm aerodynamic diameter (PM_{2.5}), than to the larger PM₁₀

particles. Because of this, the United States Environmental Protection Agency recently promulgated a new standard for PM_{2.5} of 65 $\mu\text{g}/\text{m}^3$ for a 24-h average and 15- $\mu\text{g}/\text{m}^3$ annual arithmetic mean (Federal Register 40 CFR Part 58).

Sources of PM_{2.5} are primarily combustion and conversion from precursor gases, as mechanical operations are generally inefficient at producing such fine particles. Nevertheless, agricultural operations can generate dust clouds from soil operations with size distributions that extend below 2.5 μm (Matsumura et al., 1996).

Although it is thought that the composition of particulate matter is an important determinant of health effects, many different types of fine PM have been implicated. Lipsett et al. (1995) found adverse

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effects when the primary form of PM_{10} was residential wood combustion. Gordian et al. (1995) concluded that environmental silica from geological sources increased asthma and upper respiratory illness. Aust et al. (1995) determined that iron-containing particles could cause pulmonary cell damage.

We have constructed a dust resuspension chamber to generate fugitive dust for geological source profiles and to investigate the potential of soil to emit dust in the PM_{10} size range (Carvacho et al., 2002). Details on the experimental protocol are found in the work of Carvacho et al. (2002). Using the same protocol, we have also investigated the potential of soil to emit dust in the $PM_{2.5}$ size range. The $PM_{2.5}$ is separated from the dust cloud using a $PM_{2.5}$ cyclone and is collected on Teflon filters for gravimetric and elemental analysis. The mass of dust generated from the soil sample as a function of time can be modeled by a decaying exponential function. The model parameters are related to the inherent $PM_{2.5}$ emission potential of the soil and the energy input necessary to separate the $PM_{2.5}$ from the parent material.

We have optimized the chamber operating parameters to produce results that can be related to underlying soil properties. We have tested the procedure on 44 soils spanning a range of soil textures. The chamber gives consistent and repeatable results when used with the optimized operating parameters. This paper describes the potential of geological material to emit $PM_{2.5}$ based on the 44 soils tested, and also compares these results to the earlier results for PM_{10} .

2. Materials and methods

Since we want to measure the maximum potential for soil to emit particles, we oven dry the soil at 105 °C overnight prior to performing the tests. Approximately 1 g of dry soil material sieved to the size fraction of 0–75 μm is placed in the fluidizing bed dust resuspension chamber, which is then sealed. An aluminum tube of 1.0-cm diameter connects the end of the dust suspension chamber to the inside of the dust collection chamber.

A measured volume of air (3.5 lpm for 15 s) is forced through the soil sample at the base of the fluidizing bed. The upward velocity of the air stream is sufficient to suspend dust particles of $\sim 50\text{-}\mu m$ aerodynamic diameter. These particles are carried out of the resuspension chamber and into the collection chamber as shown in Fig. 1. The particles are then collected on a 47-mm Teflon filter after passing through an AIHL-design $PM_{2.5}$ cyclone.

The 15-s “puff” of dust is collected for 15 min onto a single Teflon membrane filter. We repeat this procedure using the same sample of the soil until it is depleted of $PM_{2.5}$ material. The procedure we have established is to repeat the dust “puff” 10 times, collecting the first puff on a single filter, the second and third puffs on a second filter, the next three puffs on a third filter, and the last four puffs on a fourth filter. The IMPROVE sampler used for this purpose contains filter holders for four filters, and allows us to switch filters instantaneously without stopping the procedure.

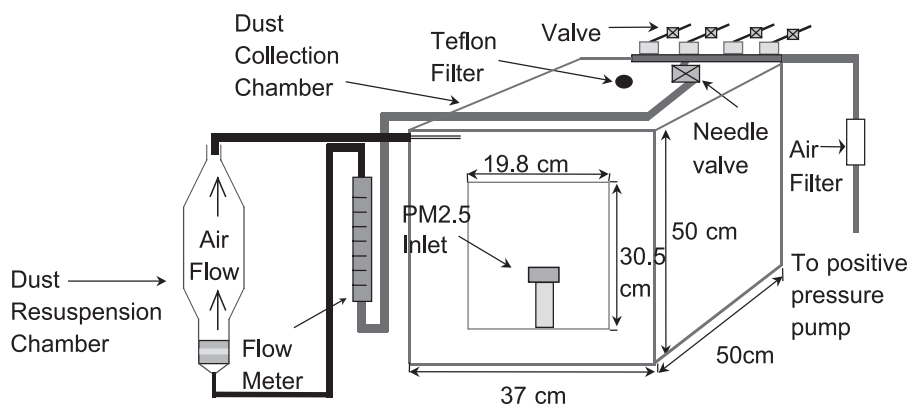


Fig. 1. Schematic of the CNL dust resuspension and collection chamber.

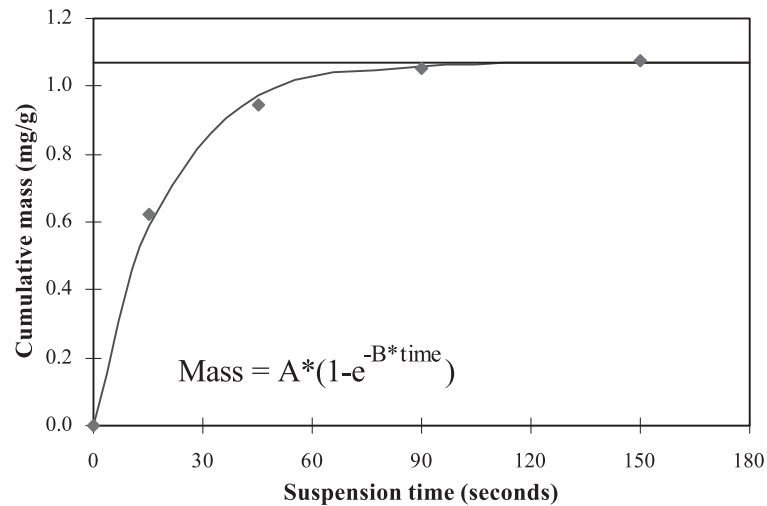


Fig. 2. Curve fit method for PM_{10} or $PM_{2.5}$ index.

The $PM_{2.5}$ (or PM_{10}) Index is calculated by fitting the cumulative mass (CM) collected as a function of suspension time t to the equation $CM = A*(1 - e^{-Bt})$. The curve fit is shown graphically in Fig. 2. The time parameter is the cumulative time (in seconds) of soil suspension in the fluidizing bed resuspension chamber. The parameter A is the asymptote of the decaying exponential curve and represents the PM_{10} or $PM_{2.5}$ Index. This represents the maximum amount of PM_{10} or $PM_{2.5}$ that would be

released by repeated “puffs” if disaggregation did not occur.

For this study, we collected 44 soil samples from agricultural fields, unpaved roads, paved roads, disturbed land areas, construction sites, and equipment staging areas in California’s San Joaquin Valley. These soils spanned a wide range of texture, as shown in Fig. 3. Some of the agricultural soils were replicates from different parts of the same field. Generally, the unpaved road sample was collected from agricul-

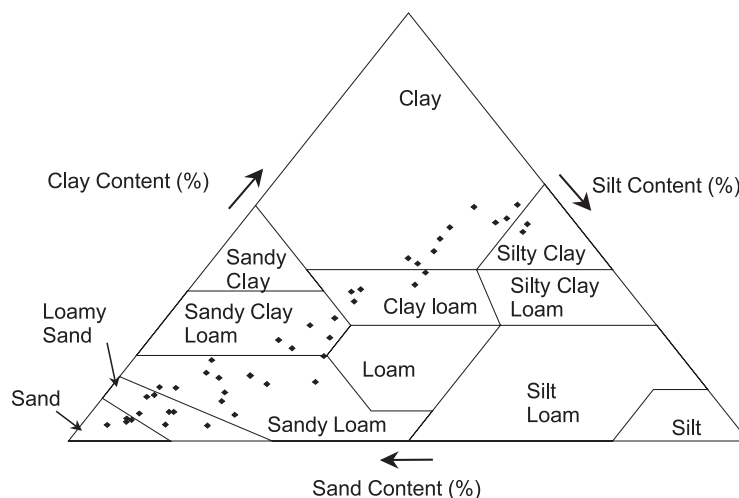


Fig. 3. Distribution of San Joaquin Valley soil textures collected and analyzed.

tural roads adjacent to the field where the crop soil sample was collected.

We measured the sand, silt, and clay content of each soil by wet sieving and gravimetric pipette suspension. This represents the soil particle size distribution for completely disaggregated soil, and is the standard measurement used by soil scientists for soil texture.

3. Results and discussions

Fig. 4 shows the relationship between the $PM_{2.5}$ Index and the standard soil texture parameters sand,

silt, and clay. The $PM_{2.5}$ Index is plotted for the 0–75- μm fraction of dry-sieved soil; recall that the index is the maximum amount of $PM_{2.5}$ dust that is generated from 1 g of soil material. Table 1 summarizes the regression statistics for the $PM_{2.5}$ Index on the soil texture parameters and compares them to the regression statistics found earlier for the PM_{10} Index.

As we found earlier with the PM_{10} Index, the $PM_{2.5}$ Index is correlated better to the sand (Fig. 4a) and clay (Fig. 4c) content of the soil than to the silt content (Fig. 4b). This may be related, in part, to the measurement method; the sand and clay contents are measured directly and the silt content is obtained by

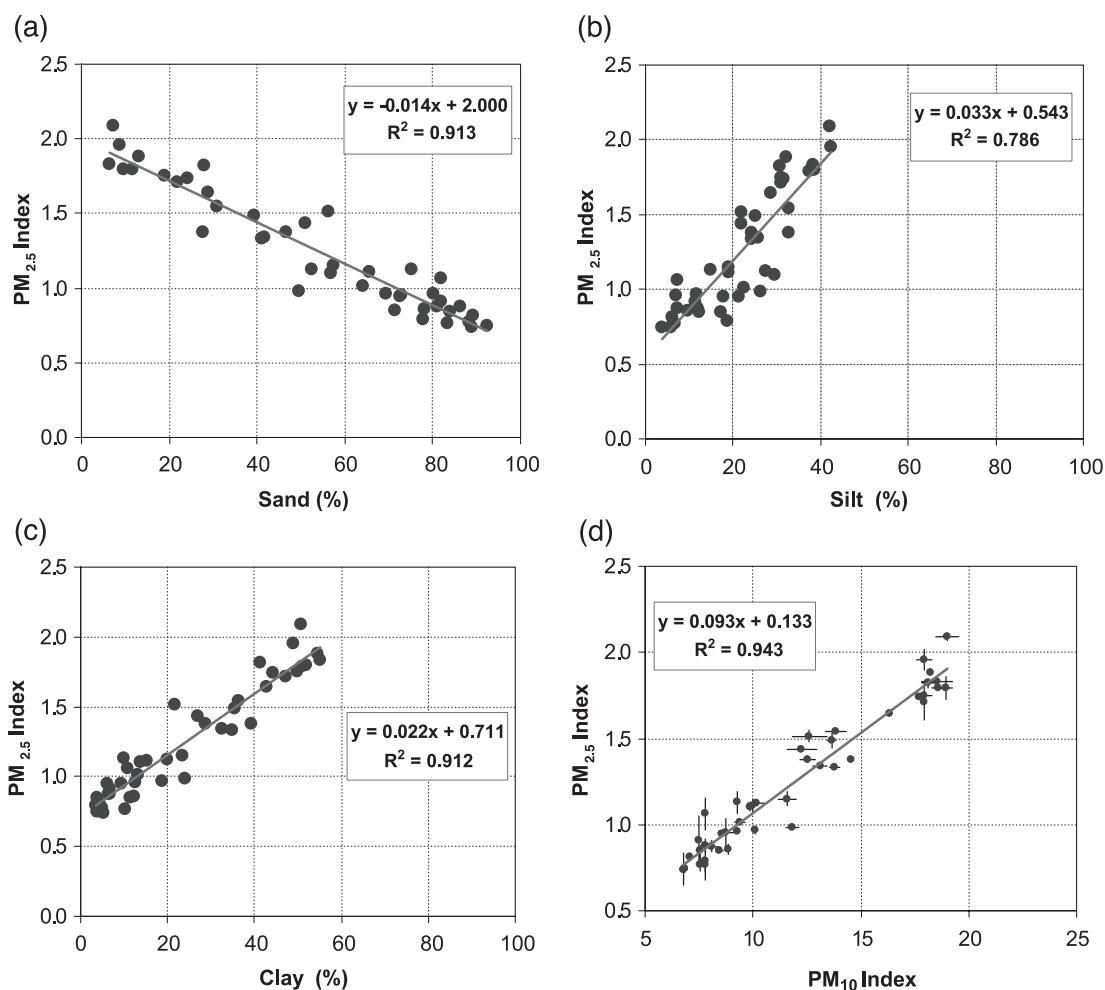


Fig. 4. The relationship between the $PM_{2.5}$ index and soil texture parameters. (a) Relationship to percent sand (particles 50–2000 μm), (b) relationship to percent silt (particles 2–50 μm), (c) relationship to percent clay (particles < 2 μm), and (d) relationship between PM_{10} index and $PM_{2.5}$ index.

Table 1
Regression statistics for PM₁₀ and PM_{2.5} index on soil texture measurements

Soil texture	PM ₁₀			PM _{2.5}		
	Slope	Intercept	r^2	Slope	Intercept	r^2
Sand	−0.148	19.937	0.959	−0.014	2.000	0.913
Silt	0.341	4.540	0.801	0.033	0.543	0.786
Clay	0.237	6.175	0.973	0.022	0.711	0.912

subtracting the sand and clay content from 100%. In any case, there is a very good correlation ($r^2 > 0.9$) between the PM_{2.5} Index and the clay or sand fractions. The predicted PM_{2.5} Index can be improved by taking the average of the index predicted by the sand and the clay content. Thus, the procedure we have adopted for predicting the PM₁₀ or PM_{2.5} Index is to average the two quantities calculated from the regression equations for the sand and clay content (Fig. 4 plots the measured PM₁₀ or PM_{2.5} Index, not the calculated one).

The current procedure recommended by the U.S. EPA in AP-42 to predict dust emission from soil preparation operations relies on a measure of the soil dry silt content (U.S. EPA, 1995). The dry silt content is the soil fraction measured by dry sieving that is less than 75 μm , i.e. the fraction of soil that passes through a 75- μm sieve. It is not the same as the silt content measured by wet sieving and gravimetric pipette suspension (the latter is plotted in Fig. 4b). The dry silt content is not readily available for a given soil, so the recommended AP-42 procedure for estimating soil dust emissions includes a provision for using a default value of 18%.

We have found that both the PM₁₀ and PM_{2.5} Index are better correlated to the soil texture measures than to the dry silt content. The dry silt content ranged from 2% to 48% for the soils used in this study, and would not be well represented by a default value of 18%. We expect the PM₁₀ and PM_{2.5} Indexes to be more useful parameters to use in emission calculations, once they are linked to actual emission measurements and we have accounted for other key determinants of emission. These determinants may include soil moisture, ambient relative humidity, and type of soil operation.

Fig. 4d shows the relationship between the PM₁₀ Index and the PM_{2.5} Index. Note that each index was

measured twice; Fig. 4d plots the average of the two measurements and shows the standard deviation as the error bars. The ratio is quite reproducible; the mean relative standard deviation for the PM₁₀ Index was 1.8%, and for the PM_{2.5} Index was 3.3%. The maximum relative standard deviation was 6.4% and 14.7% for the PM₁₀ Index and the PM_{2.5} Index, respectively. The average ratio of PM_{2.5} Index to PM₁₀ Index is 0.11 ± 0.01 [range 0.08–0.14]. Matsumura et al. (1996) found a PM_{2.5}/PM₁₀ ratio of 0.18 ± 0.12 for dust measurements at the downwind edge of a field during harvesting and land preparation activities. Thus, it appears that the PM₁₀ or PM_{2.5} Index is a useful measure of the potential of soil to emit dust, and will ultimately provide a method to estimate the emissions of dust based on readily available or easily obtained information.

4. Summary and conclusions

We have established a method to measure the potential of soil to emit fugitive dust in the PM₁₀ or PM_{2.5} size range. The method is repeatable, and provides an index of PM₁₀ or PM_{2.5} dust that is highly correlated to the soil texture. The ratio of PM_{2.5} to PM₁₀ produced by this method is similar to field observations of PM_{2.5} and PM₁₀ concentrations downwind of agricultural operations in the San Joaquin Valley of California. The PM_{2.5} or PM₁₀ Index will be a more useful parameter to estimate the potential of a soil to emit fugitive dust than the currently used dry silt content of soil. Research is currently underway to relate the PM₁₀ and PM_{2.5} Index to measured emission factors, accounting for soil moisture and type of agricultural operation, so that a more reliable predictive equation can be developed for agricultural practices.

References

- Aust, A.E., Smith, K.R., Ball, J.C., 1995. Iron in airborne particulates may be important in lung damage. *Particulate Matter: Health and Regulatory Issues*. Proceedings of an International Specialty Conference. A&WMA, Pittsburgh, PA, pp. 169–176.
- Carvacho, O.F., Ashbaugh, L.L., Brown, M.S., Flocchini, R.G., 2002. Relationship between San Joaquin Valley soil texture and PM₁₀ emission potential using the UC Davis dust resus-

- pension test chamber. Transactions of the ASAE 44 (6), 1603–1608.
- Gordian, M.E., Morris, S., Özkaynak, H., Xue, J., Spengler, J., 1995. Particulate air pollution and respiratory disease in Anchorage, Alaska. Particulate Matter: Health and Regulatory Issues. Proceedings of an International Specialty Conference. A&WMA, Pittsburgh, PA, pp. 143–166.
- Lipsett, M., Hurley, S., Ostro, B., 1995. Winter air pollution and emergency department visits for asthma in the San Francisco Bay Area. Particulate Matter: Health and Regulatory Issues. Proceedings of an International Specialty Conference. A&WMA, Pittsburgh, PA, pp. 105–113.
- Matsumura, R.T., Ashbaugh, L., James, T., Carvacho, O., Flocchini, R., 1996. Size distribution of PM₁₀ soil dust emissions from harvesting crops. Proceedings of the International Conference on Air Pollution from Agricultural Operations, Kansas City, Midwest Plan Service, Aines, IA, pp. 147–154.
- Pope, C.A., Dockery, D.W., 1998. Epidemiology of particle effects. In: Lort, S. (Ed.), Air Pollution and Health. Academic Press, San Diego, CA, pp. 673–705.
- Pope, C.A., Dockery, D.W., Schwartz, J., 1995. Review of epidemiological evidence of health effects of particulate air pollution. *Inhalation Toxicology* 7, 1–18.
- U.S. EPA, 1995. AP-42, Compilation of Air Pollutant Emission Factors, Washington, DC, vol. I. 5th ed.
- U.S. EPA, 1996. Air Quality Criteria for Particulate Matter. Office of Research and Development, Research Triangle Park, NC. EPA/600/P-95/001aF.